

JNTUK R23 REGULATION · SEMESTER I

Basic Civil & Mechanical Engineering

Unit I – Basics of Civil Engineering

Subject Code:

20BS1101

University:

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Regulation:

R23

Semester:

I – B.Tech

- ◆ Introduction to Civil Engineering & Role of Civil Engineers
- ◆ Disciplines: Structural, Geotechnical, Transportation
- ◆ Hydraulics & Water Resources Engineering
- ◆ Environmental Engineering
- ◆ Building Construction & Planning
- ◆ Construction Materials: Cement, Aggregates, Bricks, Concrete, Steel
- ◆ Prefabricated Construction Techniques
- ◆ Q&A Section · Quick Revision Sheet · References

1. Introduction to Civil Engineering

1.1 Definition

Civil Engineering is the oldest and broadest branch of engineering that deals with the design, construction, maintenance, and management of infrastructure essential for the functioning of modern civilisation. It encompasses built environments including roads, bridges, canals, dams, airports, sewage systems, pipelines, and buildings.

Etymology: The term "Civil Engineering" was first coined by **John Smeaton** (1761) to distinguish it from military engineering. The word "civil" originates from the Latin *civilis*, meaning "of or relating to citizens."

1.2 Historical Evolution

Civil engineering has existed since antiquity. Early humans built shelters, then communities developed irrigation systems, roads, and monumental structures. Key historical milestones:

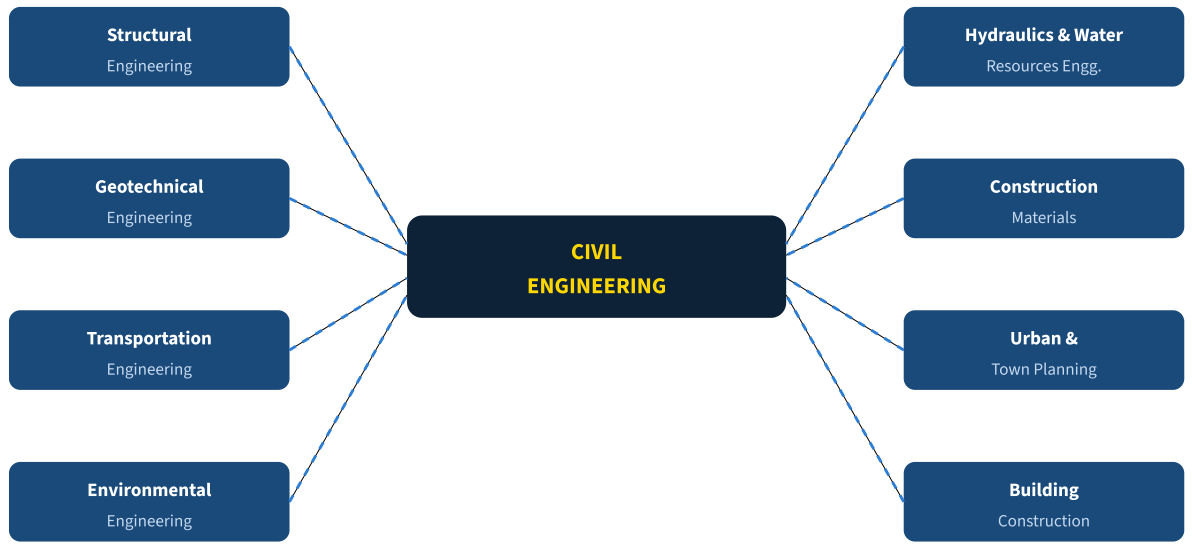
- **Ancient Era (3000 BC – 500 AD):** Construction of the Great Pyramids of Giza (Egypt), Roman aqueducts, and the Great Wall of China demonstrated early civil engineering feats.
- **Medieval Era (500–1500 AD):** Construction of cathedrals, castles, and harbour works in Europe.
- **Industrial Era (1750–1900):** Steam engines enabled rapid construction of railways, iron bridges, and urban infrastructure.
- **Modern Era (1900–present):** Reinforced concrete, steel-framed skyscrapers, highways, dams, and smart infrastructure.

1.3 Scope of Civil Engineering

The scope of civil engineering is extremely wide, covering:

Domain	Examples
Infrastructure	Roads, railways, airports, ports
Structural Works	Buildings, bridges, towers, dams
Water Resources	Dams, canals, pipelines, reservoirs
Geotechnical	Foundation engineering, slope stability
Environmental	Water treatment, sewage disposal, pollution control
Urban Planning	Town planning, land use, zoning

Fig. 1.1 – Civil Engineering Disciplines Flow Diagram



2. Role of Civil Engineers in Society

2.1 Definition & Overview

Civil Engineers are the architects of society's physical framework. They are responsible for planning, designing, constructing, operating, and maintaining infrastructure systems that underpin the quality of life of populations across the world. Without civil engineers, modern civilisation as we know it would be impossible.

2.2 Key Roles & Responsibilities

#	Role	Description
1	Planner	Plan infrastructure considering population growth, geography, and resources
2	Designer	Design safe and economical structures, roads, dams, and systems
3	Constructor	Supervise and manage construction activities at site level
4	Researcher	Develop new materials, construction methods, and technology
5	Manager	Manage projects, budgets, timelines, and teams
6	Consultant	Provide technical advice to government and private agencies
7	Environmental Guardian	Ensure sustainable and eco-friendly construction practices
8	Disaster Manager	Design earthquake-resistant, flood-resistant, and resilient structures

2.3 Contributions to Society

✓ Societal Contributions • Provide safe drinking water through treatment plants • Construct hospitals, schools, and public buildings • Build transport networks connecting communities • Design sewage systems for public health • Manage rivers and flood plains for safety • Construct renewable energy infrastructure	🏗️ Infrastructure Impact • Bridges connect geographically isolated communities • Dams provide water for irrigation and power • Highways facilitate economic growth • Airports & ports enable global trade • Urban planning improves quality of life • Disaster-resilient design saves lives
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2.4 Professional Ethics of a Civil Engineer

Civil Engineers must adhere to the **Code of Ethics** prescribed by professional bodies such as *Institution of Engineers India (IEI)* and *ASCE (American Society of Civil Engineers)*. Core ethical principles include:

- Public safety and welfare shall be the paramount concern
- Engineers shall practise only within their area of competence

- Engineers shall act in a manner that enhances public trust in the profession
- Engineers shall not engage in corrupt practices or conflicts of interest
- Sustainability and environmental stewardship are professional obligations

2.5 Civil Engineering in Modern India

India's rapid development has made civil engineers critical stakeholders. Key national projects include:

- **Sardar Sarovar Dam** – One of the world's largest gravity dams on River Narmada
- **Delhi Metro** – Underground and elevated rail network across 390 km
- **Mumbai–Pune Expressway** – High-speed road cutting through the Western Ghats
- **Atal Tunnel (Rohtang)** – World's longest high-altitude road tunnel at 9.02 km
- **Smart Cities Mission** – Urban infrastructure development across 100 cities

Important Note for Exam: The role of a civil engineer can be classified under *Planning* → *Design* → *Construction* → *Maintenance* → *Management*. Remember this sequence for 5-mark answers.

3. Disciplines of Civil Engineering

3.1 Structural Engineering

Definition

Structural Engineering is the sub-discipline of civil engineering that deals with the analysis and design of structures that support or resist loads. It ensures that structures are strong, stable, rigid, and durable throughout their service life.

Working Principle

1. Identify all loads acting on the structure (dead load, live load, wind load, seismic load).
2. Analyse the internal forces (bending moment, shear force, axial force) using structural analysis methods.
3. Design structural members (beams, columns, slabs, footings) to safely carry these internal forces.
4. Select appropriate materials (RCC, steel, timber) based on design requirements.
5. Prepare construction drawings and specifications for execution.
6. Monitor and inspect during construction to ensure code compliance.

KEY FORMULAE – STRUCTURAL ENGINEERING

Stress (σ) = Force (F) / Area (A) [N/mm²]

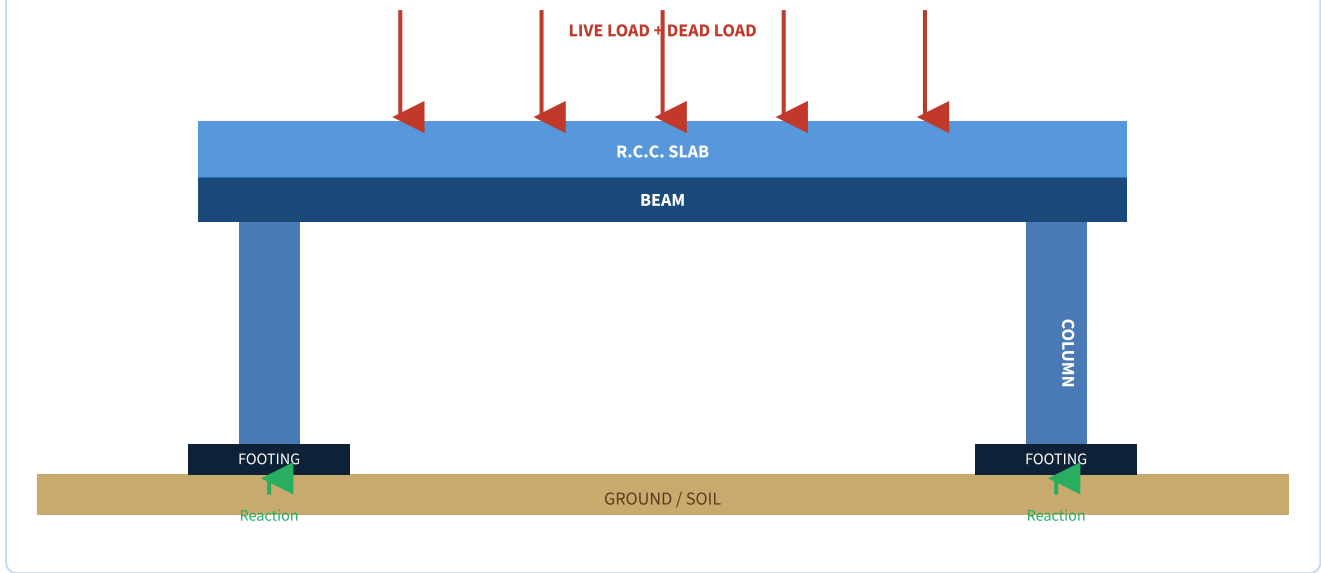
Strain (ϵ) = Change in Length (ΔL) / Original Length (L)

Young's Modulus (E) = Stress / Strain [N/mm²]

Bending Moment (M) = Force × Distance [kN·m]

Factor of Safety = Ultimate Strength / Working Stress

Fig. 3.1 – Load Distribution in a Structure



Scope & Applications

✓ Advantages

- Ensures safety of human lives
- Optimises material usage
- Enables construction of complex forms
- Provides long service life

✗ Limitations

- Requires highly skilled analysis
- Expensive software and testing
- Unforeseen loads can cause failure
- Maintenance-intensive over time

Applications: Bridges, High-rise buildings, Dams, Industrial sheds, Towers, Nuclear containment structures, Offshore platforms.

3.2 Geotechnical Engineering

Definition

Geotechnical Engineering (also called Soil Mechanics and Foundation Engineering) is the branch that deals with the engineering behaviour of earth materials — soils and rocks — and their application to the design of foundations, retaining walls, slopes, embankments, and underground structures.

Working Principle

1. Conduct **site investigation** including borehole drilling, Standard Penetration Tests (SPT), and soil sampling.
2. Perform **laboratory tests** to determine grain size distribution, Atterberg limits, shear strength, and compressibility.
3. Classify soil (using IS 1498 / USCS) and determine bearing capacity of the soil stratum.
4. Design an appropriate **foundation type** (shallow or deep) based on soil bearing capacity and structural loads.
5. Assess slope stability, liquefaction potential, and settlement analysis.
6. Provide recommendations for soil improvement techniques if native soil is inadequate.

KEY FORMULAE – GEOTECHNICAL ENGINEERING

Ultimate Bearing Capacity (Terzaghi): $q_u = cN_c + qN_q + 0.5 \cdot \gamma \cdot B \cdot N_\gamma$

Void Ratio (e) = Volume of Voids / Volume of Solids

Porosity (n) = Volume of Voids / Total Volume $\times 100\%$

Degree of Saturation (S) = Volume of Water / Volume of Voids $\times 100\%$

Shear Strength (Coulomb): $\tau = c + \sigma \cdot \tan(\phi)$

Fig. 3.2 – Soil Profile / Soil Layers



✓ Advantages

- Prevents foundation failures
- Optimises foundation design
- Ensures slope stability
- Reduces settlement damage

✗ Limitations

- Soil variability is unpredictable
- Extensive investigation is costly
- Ground conditions can change with weather

Applications: Foundation design, Embankments, Retaining walls, Tunnels, Earthen dams, Landfills, Offshore platforms.

3.3 Transportation Engineering

Definition

Transportation Engineering is the application of technology and scientific principles to the planning, functional design, operation, and management of facilities for any mode of transportation in order to provide safe, rapid, comfortable, convenient, economical, and environmentally compatible movement of people and goods.

Working Principle – Road Design Process

1. Traffic survey and prediction of future traffic volume (in PCU – Passenger Car Unit).
2. Route selection considering terrain, existing land use, and environmental impact.
3. Geometric design: horizontal alignment (curves), vertical alignment (gradients), and cross-sections.
4. Pavement design: select pavement type (flexible / rigid), thickness of layers based on soil CBR and traffic load.
5. Drainage design to ensure water does not damage the road structure.
6. Bituminous paving (for flexible) or concrete paving (for rigid) and surface finishing.
7. Road furniture: signs, road markings, guard rails, lighting installation.

KEY FORMULAE – TRANSPORTATION ENGINEERING

Design Speed (V) linked to Radius of Curve (R): $R = V^2 / (127(e + f))$

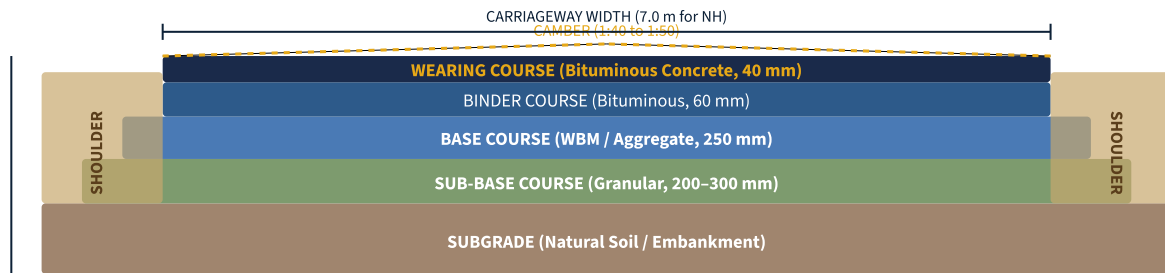
Sight Distance (SSD) = $0.278 \cdot V \cdot t + V^2 / (254 \cdot f)$

Volume (V) = Flow Rate × Time [PCU/hour]

Degree of Curve (D) = $1719 / R$ (for 30 m chord)

CBR (California Bearing Ratio) governs pavement thickness

Fig. 3.3 – Typical Road Cross-Section (Flexible Pavement)



✓ Advantages

- Connects remote regions
- Reduces travel time and cost
- Boosts economic development
- Facilitates emergency services

✗ Limitations

- High construction and maintenance cost
- Land acquisition issues
- Traffic congestion in urban areas
- Environmental impact (air, noise)

Applications: National/State Highways, Urban Roads, Bridges & Flyovers, Airports, Harbours, Railways, Tunnels.

3.4 Hydraulics & Water Resources Engineering

Definition

Hydraulics is the branch of fluid mechanics that deals with the behaviour of water and other liquids in motion or at rest. **Water Resources Engineering** extends this to the planning, development, and management of water resources for beneficial use including irrigation, water supply, flood control, and hydropower.

Key Concepts

KEY FORMULAE – HYDRAULICS & WATER RESOURCES

Continuity Equation: $Q = A_1V_1 = A_2V_2$

Bernoulli's Equation: $P/\rho g + V^2/2g + Z = \text{Constant}$

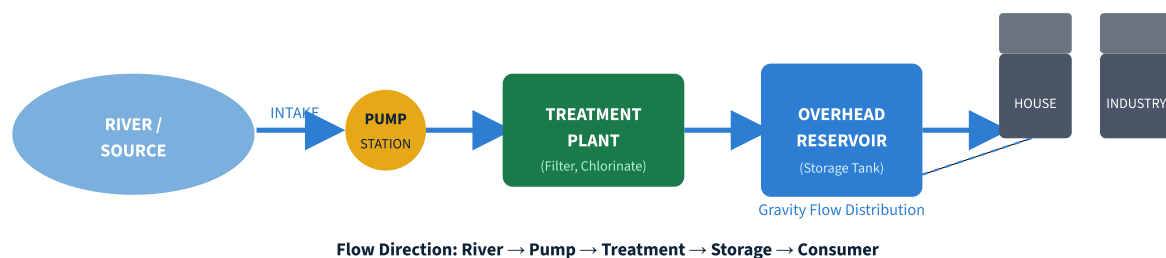
Manning's Formula: $V = (1/n) \cdot R^{(2/3)} \cdot S^{(1/2)}$

Darcy's Law (Seepage): $q = k \cdot i \cdot A$

Flow Rate: $Q = A \times V \text{ [m}^3/\text{s]}$

Reynolds Number: $Re = \rho V D / \mu$ ($Re < 2000$: Laminar; $Re > 4000$: Turbulent)

Fig. 3.4 – Water Distribution / Hydraulic System



✓ Advantages

- Ensures safe drinking water supply
- Irrigation for agriculture
- Hydropower generation
- Flood control and river regulation

✗ Limitations

- Large dams displace communities
- Sedimentation reduces reservoir capacity
- Drought affects availability
- Water pollution challenges

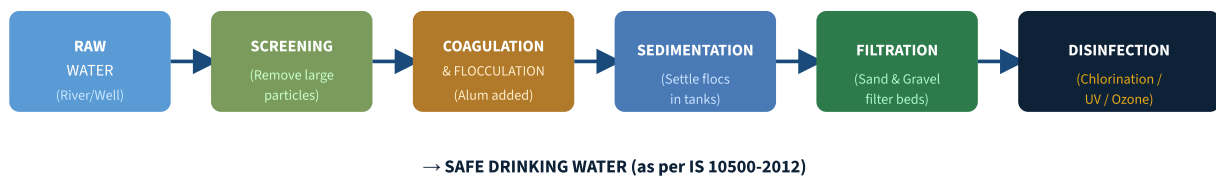
Applications: Dams and reservoirs, Irrigation canals, Water supply schemes, Flood embankments, Hydroelectric power plants, Storm water drainage.

3.5 Environmental Engineering

Definition

Environmental Engineering applies science and engineering principles to protect and improve the quality of the environment — air, water, and land — for the health, safety, and well-being of humans, and also to improve environmental quality for ecological preservation.

Fig. 3.5 – Water Treatment Process



✓ Advantages

- Provides safe drinking water
- Controls disease outbreaks
- Reduces air and land pollution
- Improves ecosystem health

✗ Limitations

- High capital investment required
- Treatment plant operation is complex
- Industrial effluents difficult to treat

Applications: Water & wastewater treatment plants, Air pollution control, Solid waste management, Environmental impact assessments, Noise control, Landfill design.

4. Building Construction & Planning

4.1 Definition

Building Construction is the process of assembling and erecting a building from raw materials into a functional structure. **Building Planning** involves arranging the various spaces and elements of a building to fulfil the functional, aesthetic, and technical requirements of the occupants.

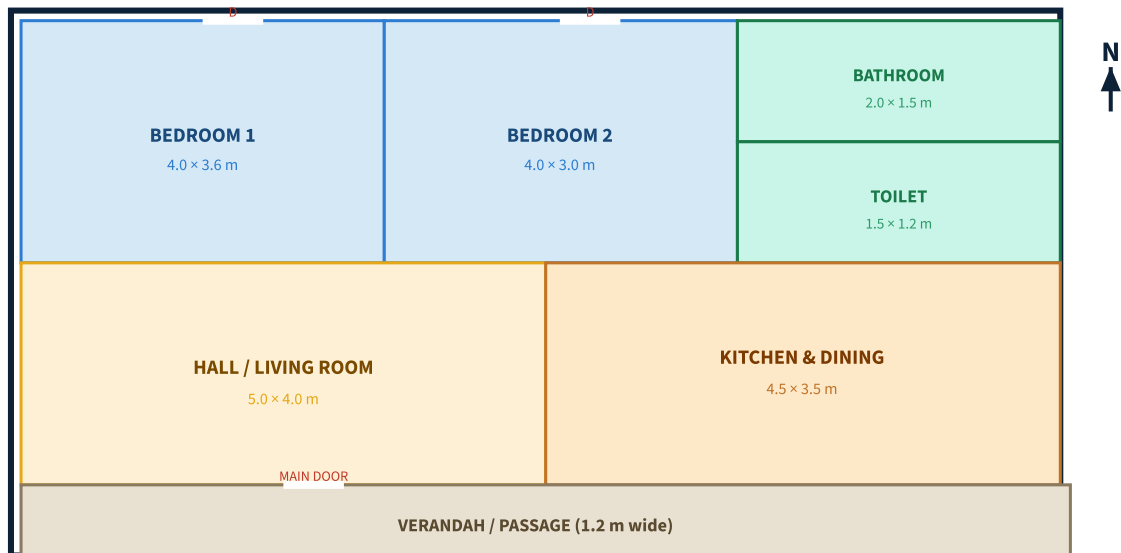
4.2 Principles of Building Planning

- **Aspect:** Positioning of rooms to receive optimum sunlight and ventilation (South-facing for India)
- **Prospect:** Pleasant view from the building (avoiding walls, industrial areas)
- **Privacy:** No direct visual access from outside into bedrooms and bathrooms
- **Furniture Requirements:** Adequate space for proper furniture placement
- **Grouping:** Rooms of similar function placed together (service, living, sleeping zones)
- **Circulation:** Efficient movement paths: corridors, staircases, passages
- **Sanitation:** Proper drainage, sewage, and ventilation systems
- **Elegance:** Aesthetic appearance of the building's exterior and interior
- **Economy:** Minimum cost without compromising quality and safety
- **Flexibility:** Design adaptable for future changes in use

4.3 Components of a Building

Component	Function	Material Used
Foundation	Transfer loads to soil	RCC, stone, brick
Plinth	Raise floor above ground, prevent moisture	Brick, concrete
Walls	Enclose space, carry loads, provide insulation	Brick, block, stone
Columns	Carry vertical loads to foundation	RCC, steel
Beams	Transfer loads from slab to columns	RCC, steel
Slab	Form floor and roof	RCC
Staircase	Vertical circulation between floors	RCC, steel
Lintel	Support masonry above openings	RCC, steel
Roof	Top covering, weather protection	RCC, GI sheet, tiles
Doors & Windows	Access, light, ventilation	Timber, aluminium, uPVC
Flooring	Finished walking surface	Tiles, marble, wood
DPC	Damp Proof Course – prevents moisture rise	Bitumen, cement mortar

Fig. 4.1 – Typical Building Layout Plan (Ground Floor)



4.4 Steps in Building Construction

- 1. Site Selection & Survey** – Topographic survey, soil investigation, legal clearances
- 2. Architectural Planning** – Prepare plans, elevations, sections as per NBC 2016
- 3. Structural Design** – Column, beam, slab, foundation design as per IS 456
- 4. Approval & Permits** – Municipal/panchayat building plan approval
- 5. Site Preparation** – Clearing, levelling, layout marking with lime/pegs
- 6. Foundation Construction** – Excavation, PCC, RCC foundation, DPC
- 7. Superstructure** – Column, beam, slab construction; brickwork
- 8. Finishing Works** – Plastering, flooring, painting, electrical & plumbing
- 9. Inspection & Handover** – Quality checks, obtaining occupancy certificate

5. Construction Materials

5.1 Cement

Definition

Cement is a fine, grey, powdery binding material which, when mixed with water, undergoes a chemical reaction called *hydration* and sets and hardens, bonding together other materials like sand and aggregates. The most commonly used type is **Ordinary Portland Cement (OPC)**.

Chemical Composition of OPC

Compound	Chemical Formula	Percentage	Role
Tricalcium Silicate (C_3S)	$3CaO \cdot SiO_2$	40–50%	Responsible for early strength
Dicalcium Silicate (C_2S)	$2CaO \cdot SiO_2$	25–35%	Responsible for long-term strength
Tricalcium Aluminate (C_3A)	$3CaO \cdot Al_2O_3$	8–12%	Rapid setting, early strength
Tetracalcium Aluminoferrite (C_4AF)	$4CaO \cdot Al_2O_3 \cdot Fe_2O_3$	8–10%	Provides colour, minor strength
Gypsum	$CaSO_4 \cdot 2H_2O$	2–3%	Retards initial setting time

Fig. 5.1 – Dry Process of Cement Manufacturing



Types of Cement

Type	IS Code	Special Property	Use
OPC 33/43/53	IS 269	General purpose	Buildings, roads, bridges
PPC	IS 1489	Fly ash blended, low heat	Mass concrete, marine works
PSC (Slag)	IS 455	GGBS blended, sulphate resistant	Underground, marine structures
SRC	IS 12330	Resistant to sulphate attack	Sewers, foundations in saline soil
Rapid Hardening	IS 8041	High early strength	Emergency repairs, precast

IMPORTANT PROPERTIES – CEMENT

Standard Consistency (SC) of cement paste: approx. 26–33% water by mass

Initial Setting Time (IST): ≥ 30 minutes (OPC) | Final Setting Time (FST):
 ≤ 600 minutes

Specific Gravity of cement: 3.15

Fineness: ≥ 225 m²/kg (Blaine's Air Permeability)

Compressive Strength of OPC 53 @ 28 days: ≥ 53 N/mm²

5.2 Aggregates

Definition

Aggregates are granular materials derived from natural rocks, quarry crushing, or recycled materials, used as inert filler in cement concrete, mortar, and bituminous mixes. They typically form **60–80%** of the total volume of concrete.

Property	Fine Aggregate (FA)	Coarse Aggregate (CA)
Size	Less than 4.75 mm	Greater than 4.75 mm
Source	River sand, M-sand	Crushed stone, gravel
IS Standard	IS 383	IS 383
Specific Gravity	2.60 – 2.65	2.60 – 2.70
Fineness Modulus	2.0 – 3.5	5.5 – 8.0
Use	Filling voids, workability	Bulk & strength of concrete

5.3 Bricks

Definition

Bricks are rectangular building units made of burnt clay or other materials, used for constructing walls, pavements, and other elements. They are one of the oldest and most widely used building materials in the world.

Standard Size (IS 1077)

- Modular Brick Size (IS 1077):** 190 × 90 × 90 mm (Nominal: 200 × 100 × 100 mm including 10 mm mortar joint)
- Traditional Size:** 230 × 110 × 75 mm
- Standard Weight:** 3.5 – 4.5 kg per brick

Fig. 5.2 – Parts of a Brick & English Bond Brickwork

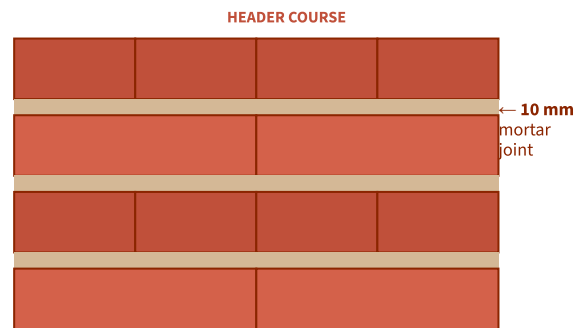
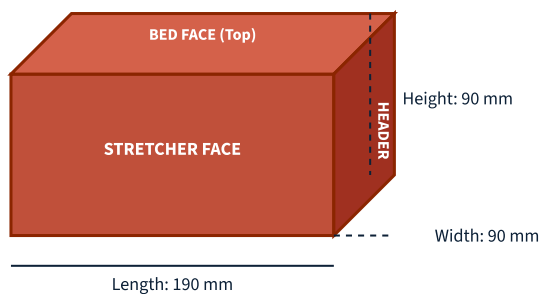


Fig: English Bond Brickwork (Alternate Header & Stretcher Courses)

Classification of Bricks (IS 1077)

Class	Compressive Strength	Water Absorption	Typical Use
Class 1 (Special)	$\geq 35 \text{ N/mm}^2$	$\leq 15\%$	Exposed & high-end work
Class 2 (First Class)	$\geq 17.5 \text{ N/mm}^2$	$\leq 20\%$	General building work
Class 3 (Second Class)	$\geq 10.5 \text{ N/mm}^2$	$\leq 22\%$	Interior & protected work
Class 4 (Third Class)	$\geq 7 \text{ N/mm}^2$	$\leq 25\%$	Temporary / non-load bearing

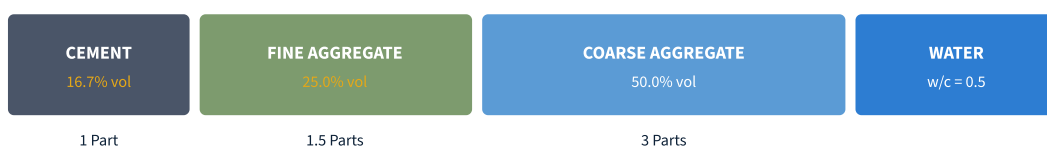
5.4 Cement Concrete

Definition

Cement Concrete is a composite construction material produced by mixing cement, fine aggregate (sand), coarse aggregate (gravel/crushed stone), water, and admixtures in appropriate proportions. When hardened, it acts like artificial stone and has excellent compressive strength.

Fig. 5.3 – Concrete Composition (M20 Grade Mix)

M20 Grade: Cement : FA : CA = 1 : 1.5 : 3 (w/c ratio ≈ 0.5)



KEY DATA – CEMENT CONCRETE

M20 Mix: Cement : FA : CA = 1 : 1.5 : 3 | w/c ratio = 0.45 – 0.50

M25 Mix: Cement : FA : CA = 1 : 1 : 2 | fck = 25 N/mm²

Density of Concrete (PCC): 2300 kg/m³ | RCC: 2500 kg/m³

Slump (workability): 25–75 mm (Beams/Slabs); 75–100 mm (Columns)

Curing Period: Minimum 28 days; 14 days for GGBS/PPC mixes

5.5 Steel in Construction

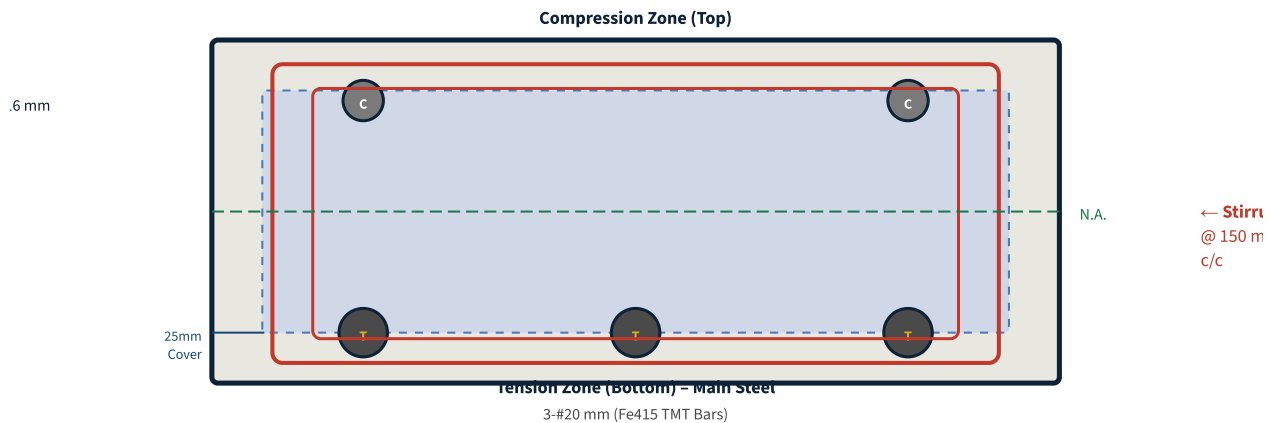
Definition

Steel is an alloy of iron with carbon (0.2–2.1% by weight) and other elements such as manganese, chromium, and nickel. In construction, steel is used as reinforcement in concrete (giving tensile strength) and as structural steel in frames, bridges, and industrial buildings.

Types of Steel Used in Construction

Type	IS Code	Yield Strength	Form	Use
Mild Steel (MS)	IS 432	250 N/mm ²	Plain bars	Light structures, links
HYSD (Fe415)	IS 1786	415 N/mm ²	Deformed (TMT) bars	RCC beams, slabs, columns
HYSD (Fe500)	IS 1786	500 N/mm ²	TMT deformed bars	Heavy RCC structures
Fe550D	IS 1786	550 N/mm ²	TMT high ductility	Seismic zones
Structural Steel	IS 2062	250 N/mm ²	Plates, angles, channels	Bridges, industrial frames

Fig. 5.4 – Steel Reinforcement in RCC Beam (Cross-Section)



Properties of Steel

MECHANICAL PROPERTIES OF STRUCTURAL STEEL

Young's Modulus (E) = $2 \times 10^5 \text{ N/mm}^2$ (200 GPa)

Shear Modulus (G) = $0.769 \times 10^5 \text{ N/mm}^2$

Poisson's Ratio (ν) = 0.30

Density = 7850 kg/m^3

Coefficient of Thermal Expansion = $12 \times 10^{-6} / ^\circ\text{C}$
Melting Point $\approx 1370\text{--}1520\text{ }^\circ\text{C}$

✓ **Advantages of Steel**

- High tensile and compressive strength
- Ductile — gives warning before failure
- Homogeneous and isotropic
- Can be recycled (sustainable)
- Fabricated off-site, fast construction

✗ **Disadvantages of Steel**

- Prone to corrosion (rusting)
- High cost compared to concrete
- Weak in fire (loses strength at 300°C)
- Requires protective coatings
- Buckling under compression

Applications: RCC structures (as reinforcement), steel-framed buildings, bridges, transmission towers, industrial sheds, offshore platforms, railway tracks.

5.6 Comparison of Construction Materials

Property	Cement	Brick	Concrete	Steel
Compressive Strength	53 N/mm ²	7–35 N/mm ²	15–50 N/mm ²	250–550 N/mm ²
Tensile Strength	Very low	Very low	~10% of f_c	Very high
Density	3150 kg/m ³	1800 kg/m ³	2400 kg/m ³	7850 kg/m ³
Fire Resistance	High	High	High	Low (needs protection)
Durability	High	High	Very High	Medium (corrosion issue)
Cost	Moderate	Low	Moderate	High
Workability	Requires skill	Easy	Moderate	Needs equipment

6. Prefabricated Construction Techniques

6.1 Definition

Prefabricated Construction (also called *Pre-fab* or *Industrialised Building System – IBS*) refers to the technique where building components — panels, columns, beams, slabs, modules — are manufactured at a factory under controlled conditions, then transported to the construction site and assembled.

Key Concept: In prefabricated construction, the majority of the manufacturing work is done in a *factory* (off-site), as opposed to traditional construction where most work is done at the *site* (in-situ). This significantly reduces on-site labour, time, and waste.

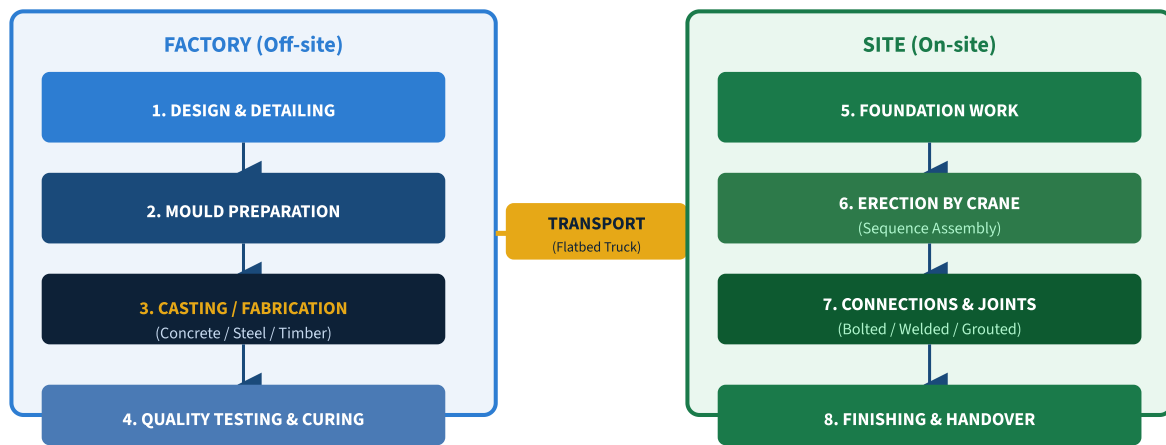
6.2 Historical Background

Prefabrication dates back to the 19th century. The **Crystal Palace** in London (1851) was one of the first large-scale prefabricated structures, made of cast iron and plate glass. In India, prefab gained momentum after Independence for mass housing projects and has grown significantly with PMAY (Pradhan Mantri Awas Yojana).

6.3 Working Principle – Manufacturing to Assembly

- 1. Design & Engineering:** Detailed structural and architectural design, with all components engineered for factory production and site assembly.
- 2. Factory Manufacturing:** Components are produced in a controlled environment with automated machinery, moulds, and quality control systems.
- 3. Quality Testing:** Each component is tested for dimensional accuracy, strength, and finish before dispatch.
- 4. Transportation:** Prefabricated units are loaded on flatbed trucks or modular transport vehicles and delivered to site. Sequence of delivery is coordinated with assembly schedule.
- 5. Site Foundation:** Conventional foundation work proceeds at site (can overlap with manufacturing).
- 6. Assembly & Erection:** Components are lifted using mobile cranes and placed as per the erection sequence. Connections are made using bolted joints, welded connections, or grouted sleeves.
- 7. Finishing & Services:** M&E (mechanical and electrical) services, glazing, and finishing works are completed. Much of M&E may also be pre-installed in factory.

Fig. 6.1 – Prefabricated Construction Process Flow



Simultaneous factory production & site preparation = Faster project delivery!

6.4 Types of Prefabricated Elements

Element	Material	Application
Precast Concrete Columns	RCC / Prestressed	Multi-storey frames
Precast Beams	RCC / PSC	Bridges, industrial buildings
Precast Hollow Core Slabs	Prestressed RCC	Floor and roof systems
Sandwich Wall Panels	Concrete + Insulation	Exterior walls
Modular Units (3D pods)	Steel / GRP	Hotels, student housing
Steel Portal Frames	Structural Steel	Industrial sheds, warehouses
Precast Staircase	RCC	Residential, commercial
Tunnel Form Slabs	RCC (in tunnel formwork)	Mass housing, dormitories

✓ Advantages of Prefabrication

- Faster construction time (30–50% quicker)
- Better quality control under factory conditions
- Reduced on-site labour requirement
- Less material wastage (10–30% reduction)
- Safer site conditions
- Works can proceed in all weather
- Reduced environmental disturbance at site
- Consistent dimensional accuracy

✗ Disadvantages of Prefabrication

- High initial investment in moulds and plant
- Transport of large units is challenging
- Limited design flexibility once mould is fixed
- Joints can be potential weak points
- Needs skilled erection crew & heavy cranes
- Not ideal for complex architectural forms
- Site access for transport vehicles required

6.5 Applications of Prefabricated Construction

- **Mass Housing:** PMAY (Pradhan Mantri Awas Yojana) projects in India use precast technology
- **Bridges:** Precast girder bridges on National Highways (PSC I-Girders)
- **Industrial Buildings:** Steel portal frame sheds with precast floor slabs
- **Railway Sleepers:** Precast prestressed concrete sleepers for tracks
- **Metro Viaducts:** Delhi Metro, Hyderabad Metro elevated sections use precast box girders
- **Defence:** Rapid deployment shelters and prefab barracks

- **Disaster Relief:** Emergency prefab shelters after earthquakes/cyclones
- **Healthcare:** Prefab operation theatres and ICU pods during COVID-19

7. Questions & Answers

Part A – 2-Mark Questions

Q1. Define Civil Engineering.

2 Marks

Definition: Civil Engineering is the branch of engineering that deals with the design, construction, maintenance, and management of physical and naturally built environments such as roads, bridges, canals, dams, airports, and buildings.

Key Point: It is one of the oldest branches of engineering and plays a vital role in building the infrastructure that supports modern civilisation. The term was first coined by **John Smeaton** in 1761 to distinguish it from military engineering.

Q2. What is the role of a Civil Engineer?

2 Marks

Civil Engineers are involved in the entire lifecycle of infrastructure: **Planning** → **Design** → **Construction** → **Operation** → **Maintenance**.

They design safe structures, manage construction projects, protect the environment, provide water supply and sanitation, build transportation networks, and ensure disaster-resilient communities. Civil engineers work across the public sector, private firms, and as independent consultants.

Q3. What is Geotechnical Engineering?

2 Marks

Geotechnical Engineering is the sub-discipline of civil engineering concerned with the engineering behaviour of earth materials. It involves the study of soil and rock mechanics and their application to the design of foundations, retaining structures, embankments, and underground works.

Key formula: *Shear Strength* (τ) = $c + \sigma \cdot \tan(\phi)$ [Coulomb's equation], where c = cohesion, σ = normal stress, ϕ = friction angle.

Q4. What is Ordinary Portland Cement (OPC)?

2 Marks

OPC is the most commonly used cement in construction, manufactured by burning limestone and clay in a rotary kiln at approximately 1450°C to produce clinker, which is then ground with gypsum.

Properties: Specific Gravity = 3.15; Initial Setting Time ≥ 30 min; Final Setting Time ≤ 600 min; 28-day compressive strength: OPC 33 ≥ 33 N/mm², OPC 53 ≥ 53 N/mm² (IS 269).

Q5. What is a brick? Give its standard size.

2 Marks

A **brick** is a rectangular building unit made from burnt clay or other materials, used in constructing walls and other masonry structures. It is one of the most durable and widely used traditional building materials.

Standard size (IS 1077): Modular brick = $190 \times 90 \times 90$ mm. With 10 mm mortar joint, the nominal size = $200 \times 100 \times 100$ mm. A good brick should have a compressive strength ≥ 3.5 N/mm² (Class 4) to ≥ 35 N/mm² (Class 1).

Q6. Define prefabricated construction.

2 Marks

Prefabricated Construction is a building technique where components (panels, columns, beams, slabs) are manufactured off-site in a factory under controlled quality conditions, then transported to the site and assembled using cranes and mechanical connections.

Advantages: Faster construction (30–50% time saving), better quality, less waste, less site labour. Used in metro viaducts, flyovers, mass housing, and industrial buildings.

Q7. Write Bernoulli's equation.

2 Marks

Bernoulli's equation states that for a steady, incompressible, frictionless flow, the sum of pressure head, velocity head, and datum head is constant along a streamline:

$$P/\rho g + V^2/2g + Z = \text{Constant (H = Total Head)}$$

where P = pressure (N/m²), ρ = density (kg/m³), g = 9.81 m/s², V = velocity (m/s), Z = datum height (m). This forms the foundation of hydraulic engineering design.

Q8. What is M20 concrete?

2 Marks

M20 refers to a concrete mix grade where M stands for Mix and 20 denotes the characteristic compressive strength of 150 mm concrete cubes at 28 days = **20 N/mm² (MPa)**.

Mix proportion (IS 456): Cement : FA : CA = 1 : 1.5 : 3 with w/c ratio ≈ 0.5 . M20 is the minimum grade for RCC structural elements as per IS 456:2000.

Part B – 5-Mark Questions

Q1. Explain the disciplines of Civil Engineering with neat diagram.

5 Marks

Civil Engineering is a broad field divided into several specialised sub-disciplines, each addressing a specific aspect of infrastructure and the built environment. The major disciplines are:

1. Structural Engineering: Deals with the design and analysis of structures (buildings, bridges, dams) to safely support loads. Key concern: strength, stability, and stiffness. Tools: IS 456, IS 800.

2. Geotechnical Engineering: Studies soil and rock behaviour for the design of foundations, embankments, retaining walls, and tunnels. Uses Terzaghi's bearing capacity theory and Coulomb's shear strength equation.

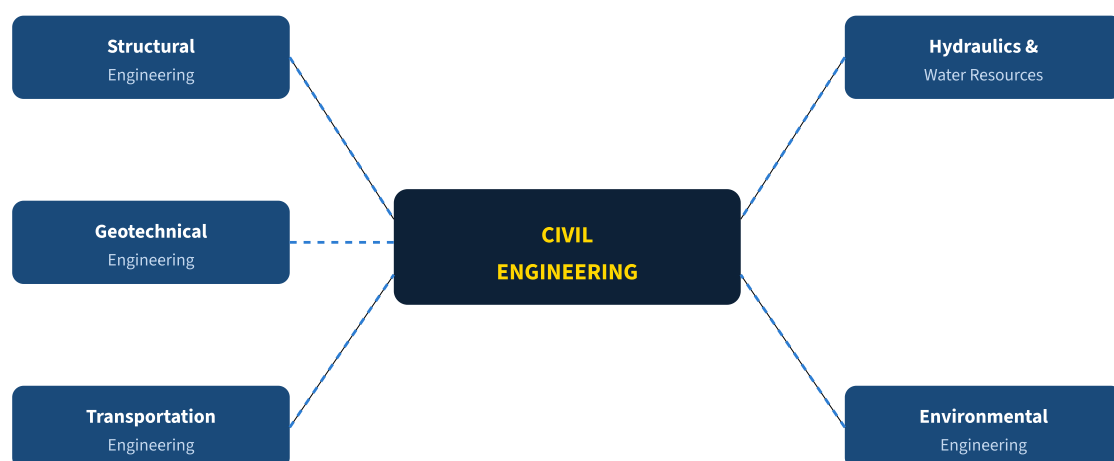
3. Transportation Engineering: Covers planning, design, operation, and maintenance of roads, railways, airports, and harbours. Focuses on geometric design, pavement design, and traffic management.

4. Hydraulics & Water Resources Engineering: Applies fluid mechanics to design dams, canals, pipelines, water supply, and drainage systems. Uses Bernoulli's equation, Manning's formula, and continuity equation.

5. Environmental Engineering: Protects and improves environmental quality by designing water and wastewater treatment plants, managing solid waste, controlling air and noise pollution, and conducting EIA studies.

Interdisciplinary Nature: Modern civil engineering projects often require expertise from multiple disciplines simultaneously — e.g., a dam project involves structural, geotechnical, hydraulic, and environmental engineering.

Fig. – Civil Engineering Disciplines (Exam Diagram)



Q2. Describe the cement manufacturing process with neat diagram.

5 Marks

Cement Manufacturing – Dry Process is the most widely used method in India. The process involves the following steps:

Step 1 – Raw Material Quarrying: Limestone (calcium carbonate – CaCO_3) is the primary raw material, quarried from limestone hills by blasting. Clay (silica, alumina) is obtained from clay pits. Both are transported to the manufacturing plant.

Step 2 – Crushing and Pre-blending: Limestone is crushed in jaw crushers to 25 mm size. Clay is separately crushed. Both are blended in correct proportions (about 80% limestone + 20% clay) to achieve the desired chemical composition.

Step 3 – Raw Mill Grinding: The blended raw material is ground in ball mills or vertical roller mills to fine powder called *Raw Meal*. The material is dried simultaneously in the dry process.

Step 4 – Preheater and Rotary Kiln: Raw meal passes through a multi-stage cyclone preheater (calcination begins at 900°C : $\text{CaCO}_3 \rightarrow \text{CaO} + \text{CO}_2$). It then enters the rotary kiln (50–100 m long) where it is heated to approximately **1450°C** , producing *clinker* (small dark balls of complex calcium silicates).

Step 5 – Clinker Cooling: Hot clinker is rapidly cooled using air in a grate cooler to below 100°C to prevent recrystallisation of minerals.

Step 6 – Grinding with Gypsum: Clinker is mixed with **3–5% gypsum** (to control setting time) and finely ground in ball mills to produce cement powder with fineness $\geq 225 \text{ m}^2/\text{kg}$.

Step 7 – Storage and Packing: Finished cement is stored in silos and packed in 50 kg bags or dispatched as bulk cement for large projects.

Q3. Explain Prefabricated Construction Techniques with advantages and applications.**5 Marks**

Prefabricated Construction (Pre-fab) is a modern construction method where building components are manufactured in a factory and assembled at the construction site. This approach is gaining increasing importance due to urbanisation, housing shortage, and the need for faster construction.

Process:

- ① Design & Engineering: Detailed structural design with connection details, tolerance, and transport constraints considered upfront.
- ② Factory Manufacturing: Precast concrete (using moulds, reinforcement, and concrete with vibration compaction), steel fabrication, or modular unit assembly under controlled conditions.
- ③ Quality Control: Each unit is tested for strength, dimensions, and surface finish. This is far superior to in-situ construction which is weather-dependent.
- ④ Transport: Components are loaded on specialised vehicles and delivered to site in erection sequence. Heavy elements may be split into segments for transport.
- ⑤ Site Assembly: Mobile cranes (50–300 tonne) lift components and position them as per erection drawings. Connections are made using bolted splices, welded joints, or grouted sleeve couplers.

Types: Precast columns, beams, hollow core slabs, sandwich wall panels, modular 3D pods, steel portal frames, precast staircases.

Advantages: 30–50% faster construction; reduced on-site labour; consistent quality; less material waste; all-weather construction; safer work environment; economical for repetitive elements.

Disadvantages: High initial mould cost; transport challenges; requires crane access; joint details need careful design; limited design flexibility.

Applications: Metro viaducts (Delhi Metro, Hyderabad Metro), PMAY mass housing, bridges (PSC girder bridges on NH), industrial sheds, railway sleepers, disaster relief shelters, COVID-19 hospital modules.

Q4. Write about construction materials – Cement, Aggregates, Bricks, Concrete, and Steel with their properties.**5 Marks**

1. Cement (OPC): A fine grey binding material that hardens on hydration. Main compounds: C_3S (early strength), C_2S (late strength), C_3A (rapid setting), Gypsum (retarder). Specific Gravity = 3.15. IST ≥ 30 min, FST ≤ 600 min. Used in concrete, mortar, grout.

2. Aggregates: Inert granular materials forming 60–80% of concrete volume. Fine Aggregate (FA): < 4.75 mm – river sand or M-sand. Coarse Aggregate (CA): > 4.75 mm – crushed stone or gravel. Specific gravity ≈ 2.60 – 2.70 (IS 383). Properties tested: bulk density, fineness modulus, elongation index, impact value, Los Angeles abrasion value.

3. Bricks: Rectangular burnt clay units ($190 \times 90 \times 90$ mm as per IS 1077). Class 1: ≥ 35 N/mm²; Class 2: ≥ 17.5 N/mm²; water absorption $\leq 20\%$. Used in load-bearing walls, partition walls, pavement (flooring bricks). Types: fly ash bricks, sand-lime bricks, hollow bricks, engineering bricks.

4. Concrete: Composite of cement + FA + CA + water (+ admixtures). M20: 1:1.5:3, $w/c \approx 0.5$, $f_{ck} = 20 \text{ N/mm}^2$. Density $\approx 2400 \text{ kg/m}^3$. Advantages: mouldable, durable, fire-resistant, economical. Cured for minimum 28 days.

5. Steel: Iron-carbon alloy used as reinforcement (TMT bars Fe415/Fe500) or structural steel (IS 2062). Young's Modulus = 200 GPa; Density = 7850 kg/m^3 . Excellent in tension (unlike concrete). Ductile, recyclable, but susceptible to corrosion and fire damage.

Q5. Explain the principles of building planning.

5 Marks

Building planning is the process of designing a building to fulfil functional, aesthetic, economic, and technical requirements. The major principles of building planning are:

1. Aspect: The building should be oriented to receive maximum sunlight and natural ventilation. In India, rooms facing south-east receive morning sun (healthy). Living rooms face south; bedrooms face south-east.

2. Prospect: The building should have a pleasant view from its main rooms. Bedrooms should not face garbage dumps or industrial areas. The architect achieves this through careful site orientation and window placement.

3. Privacy: Bedrooms and bathrooms should not be directly visible from the road or neighbour's property. Use compound walls, screens, and trees. Internal bedroom corridors ensure privacy between family members.

4. Grouping: Rooms of related functions are grouped together. Service zone (kitchen, toilets, store) should be grouped; sleeping zone (bedrooms) grouped separately from living zone (hall, dining, verandah).

5. Circulation: Efficient movement within the building through corridors, staircases, and passages. Corridors should be minimum 0.9 m wide; staircases should be easily accessible from all rooms. Avoid long, circuitous paths.

6. Sanitation: All rooms should have adequate cross-ventilation. Kitchens and toilets should have exhaust provisions. Sewage disposal should be properly designed to prevent contamination.

7. Elegance: External appearance should be aesthetically pleasing with proportional windows, doors, and facades. Internal finishes should be attractive and durable.

8. Economy: Use minimum materials without compromising safety. Avoid wastage through careful design and planning. Standardised dimensions reduce formwork and material costs.

9. Flexibility: The building should be designed to allow future modifications such as addition of rooms or change of use. Load-bearing frames allow interior reconfiguration.

10. NBC Compliance: All planning must comply with National Building Code (NBC) 2016 and local municipal bylaws regarding setbacks, FAR, coverage, and building height.

8. Quick Revision Sheet

Use this page for last-minute revision before exams. Covers all key definitions, formulae, properties, and important points.

◆ Key Definitions

- **Civil Engineering:** Design, construction & maintenance of physical infrastructure
- **Structural Engg:** Design of structures to carry loads safely
- **Geotechnical Engg:** Engineering behaviour of soil & rock
- **Transportation Engg:** Planning & design of transport systems
- **Hydraulics:** Behaviour of water in motion or at rest
- **Environmental Engg:** Protection of environment through engineering
- **Prefabrication:** Off-site factory production of building components
- **OPC:** Ordinary Portland Cement – most used binding material

📐 Key Formulae

- **Stress** = F/A [N/mm^2]
- **Strain** = $\Delta L/L$ [dimensionless]
- **Young's Modulus E** = Stress/Strain
- **Flow Rate Q** = $A \times V$ [m^3/s]
- **Bernoulli:** $P/\rho g + V^2/2g + Z = H$
- **Manning's V** = $(1/n) \cdot R^{2/3} \cdot S^{1/2}$
- **Shear Strength τ** = $c + \sigma \cdot \tan(\phi)$
- **Reynolds No. Re** = $\rho V D / \mu$

🏠 Cement Properties (OPC)

- **Specific Gravity:** 3.15
- **IST:** ≥ 30 minutes
- **FST:** ≤ 600 minutes
- **Fineness:** $\geq 225 m^2/kg$
- **OPC 53 @ 28 days:** $\geq 53 N/mm^2$
- **Main Compounds:** C_3S, C_2S, C_3A, C_4AF
- **Gypsum:** 2–3% added to control setting
- **Kiln Temp:** $1450^\circ C$

🧱 Brick Properties (IS 1077)

- **Modular Size:** $190 \times 90 \times 90$ mm
- **With mortar joint:** $200 \times 100 \times 100$ mm
- **Class 1:** $\geq 35 N/mm^2$, $WA \leq 15\%$
- **Class 2:** $\geq 17.5 N/mm^2$, $WA \leq 20\%$
- **Class 3:** $\geq 10.5 N/mm^2$, $WA \leq 22\%$
- **Class 4:** $\geq 7.0 N/mm^2$, $WA \leq 25\%$
- **Weight:** 3.5 – 4.5 kg/brick
- **Bond types:** English, Flemish, Header

🔧 Steel Properties

- **Young's Modulus E:** $2 \times 10^5 N/mm^2$
- **Density:** $7850 kg/m^3$
- **Poisson's Ratio:** 0.30
- **Mild Steel:** $f_y = 250 N/mm^2$
- **Fe415 (HYSD):** $f_y = 415 N/mm^2$
- **Fe500 (HYSD):** $f_y = 500 N/mm^2$
- **Melting Point:** $\sim 1370-1520^\circ C$

🏢 Concrete Grades (IS 456)

- **M15:** 1:2:4 – PCC works
- **M20:** 1:1.5:3 – RCC slabs, beams (minimum for RCC)
- **M25:** 1:1:2 – columns, footings
- **M30 & above:** Design Mix – high-rise, bridges
- **w/c ratio:** 0.40–0.55 (lower = stronger)
- **Curing:** min 28 days
- **Density (RCC):** $2500 kg/m^3$

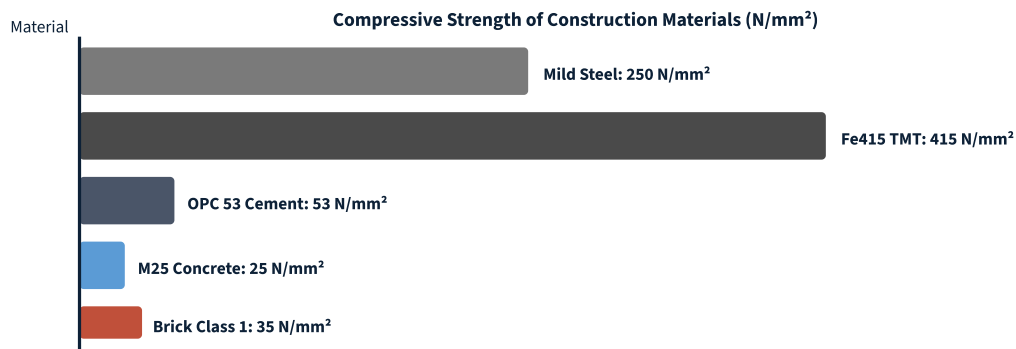
- **Standard:** IS 1786 (TMT bars)

- **Slump:** 25–100 mm (workability)

⚡ EXAM IMPORTANT POINTS:

- John Smeaton coined "Civil Engineering" in 1761 | M20 is *minimum RCC grade* (IS 456)
- Gypsum retards setting of cement (prevents flash setting)
- Prefab construction is 30–50% faster than conventional construction
- Manning's 'n' for smooth concrete channels ≈ 0.013
- Reynolds Number: <2000 Laminar; >4000 Turbulent; 2000–4000 Transition
- DPC (Damp Proof Course) is laid at plinth level to prevent moisture rise in walls
- Camber of roads: 1:40 to 1:50 (WBM); 1:50 to 1:60 (Bituminous)
- IS 10500-2012 governs drinking water quality standards in India

Quick Reference – Construction Material Strength Comparison



9. Textbook References

9.1 Primary Textbooks

📖 [T1] **M.S. Palanisamy** – *Basic Civil Engineering*, Tata McGraw Hill Publishing Co. Ltd., New Delhi. – Comprehensive coverage of civil engineering fundamentals for first-year students.

📖 [T2] **S.S. Bhavikatti** – *Introduction to Civil Engineering*, New Age International Publishers, 2022. – Covers all key topics in a structured, examination-oriented format with practical examples.

📖 [T3] **Satheesh Gopi** – *Basic Civil Engineering*, Pearson Education India, 2009. – Detailed treatment of building materials, construction techniques, and civil engineering disciplines.

9.2 Reference Books

📖 [R1] **S.K. Duggal** – *Surveying Vol I & II*, Tata McGraw Hill, New Delhi. – Standard reference for surveying, levelling, contouring, and modern survey instruments.

📖 [R2] **Santosh Kumar Garg** – *Hydrology and Water Resources Engineering*, Khanna Publishers, New Delhi. – Comprehensive reference on hydrology, runoff analysis, flood routing, and dam design.

📖 [R3] **Santosh Kumar Garg** – *Irrigation Engineering and Hydraulic Structures*, Khanna Publishers, New Delhi. – Detailed treatment of canal irrigation, weirs, barrages, and hydraulic structures.

📖 [R4] **Khanna, Justo & Veeraraghavan** – *Highway Engineering*, Nem Chand Bros., Roorkee. – Standard textbook on highway planning, geometric design, pavement engineering, and traffic engineering.

9.3 Indian Standards Codes

📖 [IS1] **IS 456 : 2000** – Plain and Reinforced Concrete – Code of Practice (Fourth Revision). Bureau of Indian Standards, New Delhi.

📖 [IS2] **IS 269 : 2015** – Specification for Ordinary Portland Cement. Bureau of Indian Standards, New Delhi.

📖 [IS3] **IS 1077 : 1992** – Common Burnt Clay Building Bricks – Specification. Bureau of Indian Standards, New Delhi.

📖 [IS4] **IS 383 : 2016** – Specification for Coarse and Fine Aggregates from Natural Sources for Concrete. BIS, New Delhi.

📖 [IS5] **IS 1786 : 2008** – Specification for High Strength Deformed Steel Bars and Wires for Concrete Reinforcement. BIS, New Delhi.

 **[IS6] IS 10500 : 2012** – Drinking Water Specification (Second Revision). Bureau of Indian Standards, New Delhi.
[Standards for potable water quality parameters]

 **[IS7] IS 800 : 2007** – General Construction in Steel – Code of Practice (Third Revision). BIS, New Delhi.

 **[IS8] National Building Code of India (NBC) 2016** – Bureau of Indian Standards, New Delhi. [Comprehensive code governing building planning, design, and construction in India]

9.4 Online Resources

 **[W1]** NPTEL Civil Engineering Courses – nptel.ac.in – Free video lectures by IIT/IISc faculty on all civil engineering subjects

 **[W2]** JNTUK Online Portal – jntuk.edu.in – Official university portal for R23 regulation syllabi and model papers

 **[W3]** Bureau of Indian Standards – bis.gov.in – Official portal for Indian Standards (IS codes)

JNTUK R23 | Basic Civil & Mechanical Engineering | Unit I Complete

This module covers 100% of the Unit I syllabus as prescribed under JNTUK R23 Regulation.
All sections include definitions, working principles, formulae, diagrams, advantages/disadvantages, and applications.

Practise all Q&A sections for examination preparation. Draw all diagrams from memory for full marks.

Best Wishes for your Examinations! 🎓